

Assignment 2: Web Scraping with Google Spreadsheet and XPath

Group 3: Kyra, Ben and Jack

*** On the submitted Google Sheet, please see both the “Good Table Live” and “URLs and Paths” Sheets! ***

Report:

To start the process of extracting the data from the HTML pages, we have to first get a list of the restaurants we want to get data from. In our case, we chose to gather data on restaurants located in the Plateau-Mont-Royal. This required us to use the URL of the webpage with the filtered results, as well as for our project, the queries restaurant names, the URL associated with each restaurant, and the star rating. To create these queries, we first used the inspect command on our computer and located the region in the code where our target information was located. From there, once the information was found in the code, we used copy XPath to help us generate a query. Once these queries were copied, we were then able to customize and merge paths to accomplish our goals. The following query was used to extract the restaurant name, their unique URL, and their star ratings (with the use of the | operator): “//div[@class='search-result']/span[contains(@id, 'searchname')]/a/text()//div[@class='search-result']/span[contains(@id, 'searchname')]/a/@href//div[@class='search-result']/span[contains(@class, 'star-rating')]/@class”. Throughout the project, we were unable to find a way to extract the exact value of the rating so as an alternative, we extracted the number of stars present and based on that information, we rounded the rating to the nearest 0.5 value.

In order for us to collect the data for the price range of each restaurant, we used a price range query to scrape the icons associated with the data such as “\$\$ & \$\$\$”. This provided us with the information on the price ranges of the restaurants formatted with the “\$” symbol, with more of them indicating the price ranges of the restaurant being higher. Since the symbols for price ranges were a broad generalization of the prices, with different interpretations of what the range should be by many people, we decided to create our own criteria for what the ranges should be. In order to do this, we used an IFS expression to evaluate multiple conditions and determine what numeric values are appropriate for every range. This ensures a clear categorization of the data which allows for consistent results throughout the dataset. The query used for this dataset was: “//*[@id="restaurant-profile"]/div[2]/div/div[1]/div[1]/div/div[1]/div/div[2]/div[1]//span[@class="restaurant-price"]/text()”.

We wanted to include information on payment methods accepted as some methods such as American Express are not always accepted at every location. In order to achieve this, we scraped the information present about the payment methods accepted on the restaurant’s unique URL. This information was listed under “Payment Methods”. “//*[@id="restaurant-profile"]/div[2]/div/div[1]/div[3]/span[@class="mb10 fs-small"]/text()” was the query used to extract the information on payment methods. The problem with this dataset is that some of the

restaurants didn't have this information listed on their website meaning, when we tried to extract this information, it returned an empty result.

Every restaurant has an address associated with them. Without this information, it would certainly be extremely hard to find it. In order to extract this information, we used the following query, along with the restaurant's unique URL. The query used was: “//*[@id="directions-tracker-2"]/span/text()”. This provided us with the civic number, the street name, and the postal code associated with the restaurant.

For the images, the process was very similar to the other datasets except an extra step was needed. We used the URLs of each restaurant and a query of “//*[@id="restaurant-profile"]/div[2]/div/div[1]/div[1]/div/div[1]/div/div[2]/div[2]/a/img/@data-src”. This provided us the link to the image. Using this link, we used the =Images() function on google sheets to get the actual images associated with the restaurant. In certain cases, some of the images we got weren't associated with the restaurant which could be the result of a lack of image present on the webpage.

In order to get the actual value of the ratings, we reformatted the data extracted previously with the TEXTJOIN function which was to keep the data standardized so it becomes easier to work with later on. We made a list of all possible combinations of the text and what it meant in terms of star ratings. We matched the result to the proper combination which in turn, gave us the star rating of the restaurant.

DOM Tree:

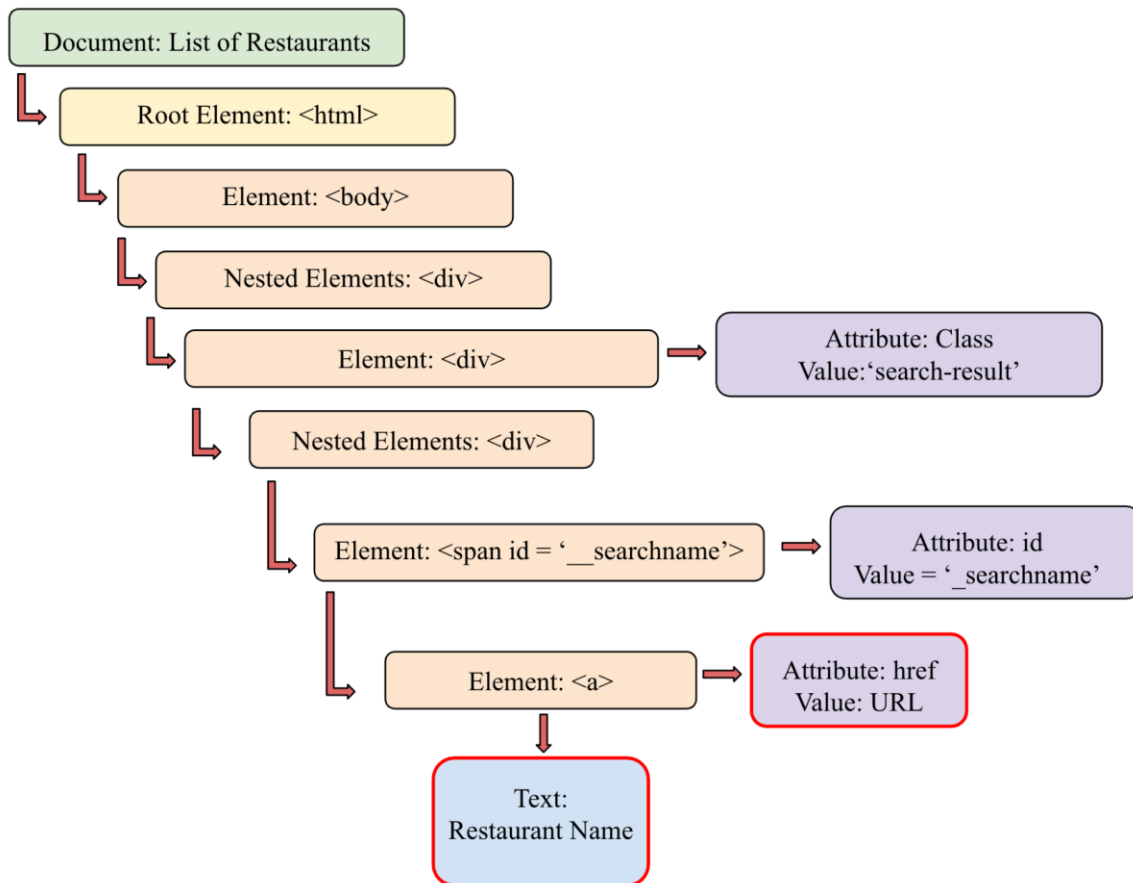


Figure 1: Document Object Model (DOM) tree.

This DOM reflects the path for scrapping the Restaurant Name and Restaurant URL. The green colour reflects the document that is being scraped, the yellow colour reflects the root element, and the orange colour reflects the various elements in which the information is found under. The purple colour reflects the attribute(s) and attribute value(s) of an element while the blue colour reflects text. The boxes outlined in red represent the items that are scraped using the specific query.

Reference XPath:

```
//div[@class='search-result']//span[contains(@id,'searchname')]/a/text() | //div[@class='search-
result']//span[contains(@id, 'searchname')]/a/@href
```

Screenshot Appendix:

Restaurant Link and Name	Price Range	Price Range Values	Payment Methods	Address	Images	Ratings	Rating Number Estimate
https://www.restomontreal.ca/resto/aiiko-sushi-montreal/4362/en/ Aiko Sushi	\$	10	Master Card, Visa, Interac	1680 Avenue du Mont-Royal E, Montreal, H2J 1	Loading...	fas fa-star search star	4.5
fas fa-star search star-rating							
fas fa-star search star-rating							
fas fa-star search star-rating							
fas fa-star search star-rating							
fas fa-star-half search star-rating							
https://www.restomontreal.ca/resto/clebard-montreal/6741/en/ Clebard	\$	10	Master Card, Visa, ATM machine	4557 Rue Saint-Denis, Montreal, H2J 2L4		fas fa-star search star	4.5
fas fa-star search star-rating							
fas fa-star search star-rating							
fas fa-star search star-rating							
fas fa-star search star-rating							
fas fa-star-half search star-rating							
https://www.restomontreal.ca/resto/saint-sushi-montreal/6343/en/ Saint Sushi	\$\$	25	Master Card, Visa, AMEX, Interac	424 Avenue Duluth E, Montreal, H2W 2M3		fas fa-star search star	5
fas fa-star search star-rating							
fas fa-star search star-rating							
fas fa-star search star-rating							
fas fa-star search star-rating							
fas fa-star-half search star-rating							
https://www.restomontreal.ca/resto/panuozzo-montreal/14632/en/ Panuozzo	\$ & \$\$	17.5	Master Card, Visa, AMEX, Interac	3820 Boulevard St-Laurent, Montreal, H2W 1X1		fas fa-star search star	4.5
fas fa-star search star-rating							
fas fa-star search star-rating							
fas fa-star search star-rating							
fas fa-star search star-rating							
fas fa-star-half search star-rating							
https://www.restomontreal.ca/resto/monsieur-b-pour-bistro-montreal/2663/en/ Monsieur B (B pour Bistro)	\$\$	25	Master Card, Visa	371 Rue Villeneuve E., Montreal, H2T 1M1		fas fa-star search star	4.5
fas fa-star search star-rating							
fas fa-star search star-rating							
fas fa-star search star-rating							
fas fa-star search star-rating							
fas fa-star-half search star-rating							
https://www.restomontreal.ca/resto/el-lounge-resto-bar-montreal/17209/en/ El Lounge Resto-Bar	\$\$ & \$\$\$	31.25	Master Card, Visa, AMEX, Interac, Mobile Payment, Contactless Payment	1735 Rue St-Denis, Montreal, H2X 3K4		fas fa-star search star	5
fas fa-star search star-rating							
fas fa-star search star-rating							
fas fa-star search star-rating							
fas fa-star search star-rating							
fas fa-star search star-rating							

Figure 2: Final Table Format

(Note: this screenshot only shows 6 restaurant examples out of 50)

Individual Reflections:

Ben:

For this assignment, I was focused on aspects relating to the XPath and scrapping processes. Therefore, I created the Google Sheet, organized the table with a standard format, and imported the data into each of the desired data categories. These categories included the unique restaurant names, subsection URLs, price range icons, monetary value estimates, payment method options, the address, logo image, and rating. To assign each restaurant with the monetary value estimate, I calculated the average of the price range for each dollar sign icon and used the IFS function to evaluate multiple conditions. As a result, the number of dollar sign icons in the

cell determined the monetary value assigned. If a combination was present (e.g. \$ & \$\$), the average between the two categories was calculated. I also was responsible for creating the DOM tree embedded in the report above and the step-by-step methodology for structuring the data table (found under the URLs and Paths Google Sheet). For the rating, it should be noted that our group had great difficulty extracting the exact rating number. Therefore, after many unsuccessful attempts and approaches, I decided to scrape the number of full and half rating stars from the general URL. Then, I was able to use the TEXTJOIN and IFS function to count the number of stars to the nearest 0.5th decimal. This method served to provide an estimate of the numerical rating value.

In the rapidly evolving world of the internet, it is essential to consider the legal regulations and potential consequences involved in gathering, analyzing, and publishing data. Particularly relevant in today's climate, Artificial Intelligence presents significant legal risks, both during its training and in its use. While the "Garbage In, Garbage Out" expression holds true for training AI, acquiring high quality input data is not always simple. As we have discussed in lecture, data scrapping is a valuable process used to extract training data, allowing access to large volumes of data while reducing human error, lowering costs, and improving data quality, performance and predictability (Jayachandran & Arni, 2023). At the same time, the price for acquiring data through scraping manifests itself through the numerous legal and ethical concerns. Legal battles between companies such as LinkedIn and hiQ Labs over the scraping of personal information from public profiles emphasize the risk associated with scrapping vulnerable data and highlight the gray-area surrounding publicly available information. In fact, the U.S Ninth Circuit of Appeals stated that hiQ Labs did not violate the Computer Fraud and Abuse Act, despite privacy concerns (Whittaker, 2022). The unease of companies accessing our personal information is certainly not something that should be ignored, and scholars argue that the future of AI "should not only be measured by its technological capabilities but also by its adherence to ethical principles and its contribution to societal good" (Jayachandran & Arni, 2023). In regards to whether publicly accessible and free-to-view sites should be available for anyone to gather and use that data for other purposes, I believe that the answer is no. While I agree that perhaps the majority of people seeking information from these pages have "good" intentions, those that lack those intentions pose grave threats to personal privacy and security. Using the term "anyone" blankets over those who wish to violate ethical boundaries, and requirements such as the Responsible AI Standard V2 mentioned by Jaychandran et al (Microsoft Responsible AI Standard v2 General Requirements, 2022). Even though the data acquisition process may be considered lawful, we have seen in this course how far the power of money and a team of lawyers can get you in the field of technology. Large corporations take advantage of the ambiguity of data scrapping laws and are quick to take data gatherers to court if there is a fear of terms and services violations. Therefore, even if data scraping may be legal, there is always a fear of legal action that large companies may take that the extractor must consider. Even though it is technically the responsibility of the users on these websites to agree to the terms and services in order to create accounts, I believe that the companies still have a sense of ethical concern to

uphold to protect the personal information of their users. While data from these websites may be useful for a variety of applications, like analyzing which products are most in demand in specific areas for a business owner seeking the best location for their store, I believe that the legal extraction of data must also be paired with ethical practices in the data scraping process and use.

Jack:

In this assignment, I helped with the data collection process, more specifically, extracting restaurant images, their addresses, and helping with trying to extract the ratings of the restaurants. Using every restaurant's unique URL, I was able to extract the address and the link for the images of the restaurants from the associated URLs. In order to extract the images from the webpages, I had to extract the link for the images first as it wasn't possible to directly extract the images and have them appear as a picture with XML. Using the image link, I was able to use the =Image() function on Google Sheets to get the actual image from the link associated with it. I helped with the process of trying to find a way to extract the ratings of the restaurants as this was the part that posed the greatest challenge for all of us in the group. For this part, I tried a variety of techniques, including using the query for the rating value indicated on the restaurant's webpage and any element where the rating was located. Unfortunately, after many different attempts such as trying by modifying the query slightly, and analyzing other html lines, I was unable to find a solution to this problem. Alternatively, as we were short on time left to finish the assignment, we finally decided to use the number of stars present to generate a rating rounded to the nearest 0.5 value.

In today's world, it's become more important than ever to consider the legal consequences of gathering data for use by AI. With the amount of information present on the internet, it can be very easy to end up scraping private information when not careful. This can lead to class action lawsuits against those scraping the data. Even if some data scraped doesn't seem like it could be classified as private information, it could be of use to eventually identify information about somebody or someone, causing concern for privacy invasion. Since the world is constantly changing, regulations on data scraping is not always up to date meaning that it's unclear what the boundaries of data scraping are. In many cases, there is data available that's public for people to see however, the website owners might require their permission for other website to scrape the information. Like the previous examples mentioned, this can lead to lawsuits against those involved with data scraping (Jayachandran & Arni, 2023). In my opinion, people should be able to gather publicly available information to use for other purposes as long as the intent is not to use it as a source of income as it might be considered unauthorized use of data by many, even going so far as being considered data theft much like plagiarism. Despite the fact that a US appeals court determined that data scraping is legal, there is still a large gray area in regards to that (Whittaker, 2022). With that ruling, I start to wonder, how can people share information that they do not want to have scraped. As of now, for me, it seems almost impossible to present information on websites that prevent scraping. Based on my experience with assignment 2, scraping the information on the ratings for instance was much harder than

expected but not impossible. It seems that for now, you could only add “layers” of protection to prevent certain levels of scraping, but that still doesn’t fully block scraping. Moreover, there is the ongoing issue of lack of transparency regarding what the data is going to be used for. This lack of transparency is especially problematic for some as it can be unknown if the data is being used for commercial purposes or for research for instance (Jayachandran & Arni, 2023).

Personally, I believe that this is one of the bigger issues that surround data scraping as the internet can be a difficult place to figure out one’s intentions with how the data is being used. I find the lack of transparency to be a problematic issue with how we perceive the use of data sources as I believe many companies are purposefully being vague about what they disclose to avoid potential consequences.

Kyra:

I contributed to this assignment by completing the tasks of HTML styling, including a) structure, b) contents, c) getting mapbox to work, and in d) code documentation. I also handled the Javascript buttons - custom icons and popups. I also was responsible for geocoding the restaurant address and parsing this data in a myriad of ways to get it from a live google sheet (scraped from RestoMontreal), to a static and usable geojson for Mapbox upload and scripting. An aspect of the process I was particularly occupied by was the photo gallery feature, which involved much trial and error and adjustment of various CSS parameters that I tested with w3 schools and GitHub, in addition to the download and upload of so many photos from the Google Sheet. I also was very occupied by the process of geocoding which I did by installing a Google Sheets extension called Geocode by Awesome Table. I manually manipulated the data in order to get into usable geojson format, via rearrangement of the spreadsheet, conversion from table to points in ArcGis Pro, exporting as a shapefile, and then uploading to geojson.io and finally importing into my newly created Mapbox style as a tileset. I particularly enjoyed adding vertical and horizontal scroll bars to my photo gallery and linked list of restaurants, the process of which taught me a lot about CSS. Having applied some basics of web development in the last assignment, I dared to experiment in making our webpage interesting and highly functional.

Along the lines of legal reflection, this assignment did give me a fair amount of anxiety around copyright law and murky legal territory. I had many thoughts about the openness of the internet and how precedents against the availability of data could make assignments like this increasingly difficult in the future. In a world where we spend so much (arguably all) of our time on the internet and in producing data, often in VGI form, it seems morally wrong that we face losing our autonomy over this domain to big companies and data brokers who operate in the interest of profit - via the privatization of our data. I thought about this particularly with certain maps that were once available on the web (and therefore for web scraping and analysis) like Snap Map and Google Timelines. These vast data banks available only to the company, limiting us to a view on the mobile app, exemplifies the loss of user mobility on the internet, a trend I only predict to spiral far and far worse. I fear the impending loss of web autonomy and scraping.

References:

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Whittaker, Z. (2022, April 18). *Web scraping is legal, US appeals court reaffirms* | TechCrunch. <https://techcrunch.com/2022/04/18/web-scraping-legal-court/>

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